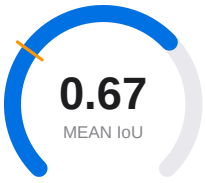
What this document is not. This is not a court-ready dossier, a legal finding, or an accusation. It records what the system computed and under which assumptions.

FINDING

419363968 is the best physical fit.

Forward replay tested 5 candidates; the runner-up fits at 0.29. A ranking by physical consistency under tested assumptions — an investigative lead for a human analyst.



MEAN IOU	VARIANTS WON	LEAD RETAINED	CANDIDATES
0.67	**5** / 6	**89%**	**5**
the fit gate is 0.3	forcing variants where it fits best	of 80 tested changes	replayed by forward replay

EVIDENCE

- Best fit under **5 of 6** forcing variants
- Ahead of the runner-up by a mean loss gap of **0.1961**
- Inside the origin area **24 h** before the image
- The nearest vessel, **419931151**, 0.0006° away, is not supported

Candidates, ranked by physical fit

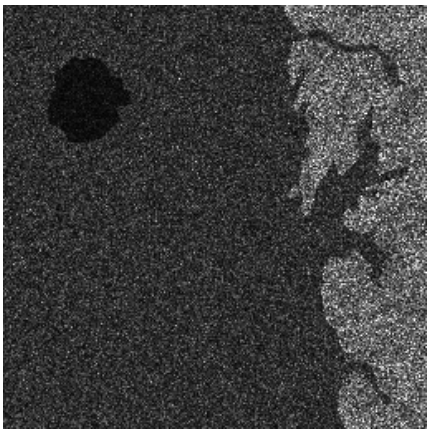
Each candidate's release replayed forward under six forcing variants and compared with the observed slick.

#	VESSEL	MEAN IoU		VARIANTS WON	RELEASE STARTS
1	419363968		0.673	5 of 6	9 Sep 2026, 22:30 UTC
2	419473598		0.288	0 of 6	9 Sep 2026, 22:30 UTC
3	419625403		0.361	0 of 6	9 Sep 2026, 11:30 UTC
4	419931151		0.210	1 of 6	10 Sep 2026, 04:00 UTC
5	419276389		0.002	0 of 6	9 Sep 2026, 10:30 UTC

The proximity baseline — the vessel nearest the slick when the image was taken — would pick **419931151**, ranked 4 by forward replay.

The scene

The radar image the slick was detected in. Radar darkness alone is not proof of oil.



Acquired	10 Sep 2026, 04:30 UTC
Sensor	synthetic SAR-like (VV proxy, single channel)
Pixel spacing	433.59 m
Data status	synthetic
Label source	generator ground truth (synthetic) — not a human or independently confirmed label
Projection	EPSG:32643

Robustness

Does the leader survive small changes to the model's assumptions?

89%
LEAD RETAINED

80 tested changes to the model's assumptions; **9** changed the leader.

The smallest change that flips it: **diffusivity multiplier 0.6** (magnitude 0.5), under which **419625403** leads.

Single-axis and two-axis perturbations only. Three-or-more-axis combinations were not tested, so absence of a flip is evidence of stability across the tested set, not a proof of stability.

Forward exposure

Where the oil is modelled to drift after the image. Reanalysis-style retrospective field — not an operational forecast.

AFTER THE IMAGE	MODELLED SPREAD	SHORELINE	FISHING GROUND	MANGROVE RESERVE
+6 h 10 Sep 2026, 10:30 UTC	478 km^2	0%	43%	0%
+12 h 10 Sep 2026, 16:30 UTC	533 km^2	0%	41%	0%
+24 h 11 Sep 2026, 04:30 UTC	586 km^2	0%	29%	0%

Traffic filtering

Every vessel in the scene, checked against where the oil could have come from over the 24 hours before the image.

5	Candidate	track data present and inside the backward envelope at some queried time
1	Unassessed	never inside where track data exists, but no track data at one or more queried times — missing evidence, NOT exoneration

Unassessed is reported separately from excluded: missing evidence is not exoneration.

Coverage-aware matching

Matching radar contacts to AIS when some AIS is hidden — tested on synthetic contacts.

RATE	VESSELS HIDDEN	AREA GOES DARK
Hidden AIS recovered	**69%** 34/49	**86%** 77/90
False alerts	**35%** 29/83	**7%** 7/97
Mistaken promotions	**30%** 12/40	**100%** 26/26

Provenance

AIS	synthetic (fictional MMSIs; no real broadcast data)
Forcing	slicktrace_core.synth_generator constant+spatial field v0.2
Licence	Synthetic data generated by this repository; released under the repository licence. No third-party data terms apply.
Generated	26 Sep 2026, 17:11 UTC
Software	Python 3.12.14 · Darwin 25.2.0 (arm64)

File integrity

Hashes verify file integrity only. They do not establish that an observation is true, that a label is correct, or that this analysis is legally sufficient.

attribution_result.json	`9d8fde0a2e513ef1207d3eb47c74c0dc1201ef7b3a69054580ecaf71fbad24e6`
coverage_result.json	`1d5336e6d34dac3c876cef149405063b51030866ce86ea3a6e939b35d7ab0b6d`
forcing.npz	`f8958ffe32637fe55198273b480cf3a0c63e50ae9ee4d619627f8e42aed43f43`
forecast_result.json	`e3da27c40cd387220e6e999a1b1910e382b2b6ee353fcfcf944e9e267bbe36fe`
fragility_result.json	`c769db1affa75514b238423607f94929ca42c6d343f5f94694d189a0dfb8bc60`
historical_mapping.json	`7d15e91a5f16841a17f6663f41ea5d08ecf83f977481add8cc708795d24fa4a2`
manifest.json	`0b5fc78586a102a03b3d8c41915d172b22471c3c6a23aa7737d2955783a28650`
mask_groundtruth.png	`180119392ca88a619e1360f8d90484c520fec896c208b0d69a7f3fb9e9a9085b`
naive_baseline_result.json	`6f46e70c8d25b4dcf7118434b0c21e5303468d228ad4ccdff48c7d10316a99e3`
next_observation_result.json	`856c2fb4d8713e077636b692d0e79928df460bcb56109aa2ccc876c748c06b5f`
scene.png	`27cd143e9af0fd36639c2fcaadb6b86180474b918bc351b62fc82ea11f1452de`

Uncertainty

- Rankings express physical consistency under tested assumptions. They are not proof of responsibility.
- Ensemble and perturbation-grid win frequencies are evidence scores, not calibrated probabilities. No calibration against held-out real outcomes exists.
- Where no candidate clears the absolute fit gate the system reports unknown source rather than naming the least-bad option.
- Synthetic ranking accuracy is not a measurement of real-world attribution accuracy.

Restated. This is not a court-ready dossier, a legal finding, or an accusation. It records what the system computed and under which assumptions.